Cancer signatures for reproducible gene expression analysis data: the computational way to achieve precision medicine

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BACKGROUND

Cancer is a complex disease, characterized by extensive genomic aberrations with an impact on gene expression and cell biological processes. In the last decade, many studies proposed **gene expression signature** as a valuable tool for understanding cancer mechanisms, defining subtypes, monitoring patient diagnosis, prognosis and therapy efficacy. However, many concerns regarding result reproducibility have been raised. One of the major concerns is the lack of a **computational implementation** of the proposed signatures, that would provide detailed signature definition and would assure reproducibility, dissemination and usability of the classifier. Yet, the vast majority of signatures are not published along with their computational code and only few of them have been implemented in a software. With the aim of providing a tool able to improve reproducibility and usability of gene expression signatures, we propose **signifinder**, an R package that collects and implements **46** expression-based signatures from cancer literature. The included signatures cover a wide range of biological processes from metabolism, programmed cell death, to morphological changes, such as quantification of epithelial or mesenchymal-like status. Collected signatures can score tumor cell characteristics, such as predict the response to therapy or the survival association, as well as quantify the microenvironmental statuses, such as hypoxia, or the activity of the immune response.

Figure 1 *- signifinder package logo.*

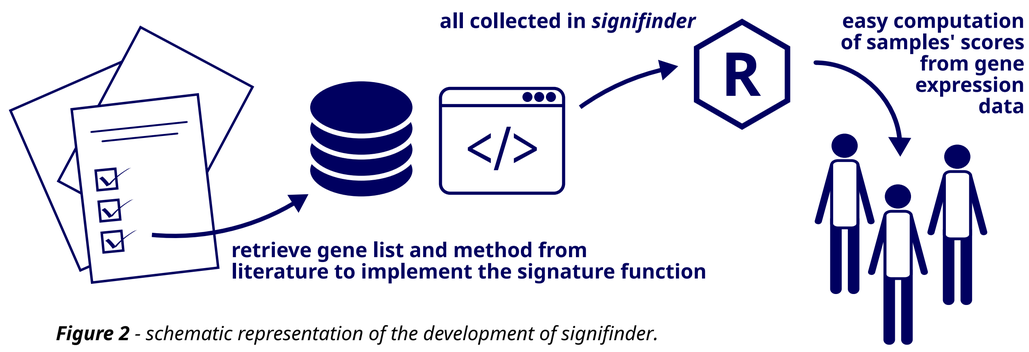

Figure 2 *- schematic representation of the development of signifinder.*

DEVELOPMENT STRATEGY

A list of signatures, covering numerous cancer hallmarks, was collected from the literature.
We established a set of **criteria for the inclusion of the signature**:

- they must be on a **cancer topic**, developed and used on cancer samples;
- they must be based on transcriptomic data from **bulk** samples;
- they must include the **gene list** and the **method** to calculate a single-sample score.

We selected 46 signatures, imported all the data needed to compute them (a specific gene set and eventually a set of coefficients to differently weight the gene contributions) inside the package and implemented dedicated **functions** that return a score designed to provide single sample predictions.

RESULTS

We developed signifinder, an R package dedicated to the computation and comparison of gene expression signatures. Overall, signifinder contains 46 signatures that cover 13 distinct cancer tissues (but also pan-tissue signatures are included) over 14 different cancer topics (**Figure 4**). We put signifinder to the test with the TCGA ovarian cancer dataset (TCGA-OVC)[1]. OVC is a disease characterized by high levels of genomic instability that impact the gene expression, causing highly variable phenotypes across patients. The transcriptional landscape of OVC is known to be differentiated in four subtypes[2] termed immunoreactive (IMR), differentiated (DIF), proliferative (PRO) and mesenchymal (MES) on the basis of the gene content and previous knowledge[3]. One of the most robust and reproducible classification of the four OVC transcriptomic subtypes is the one implemented by Chen and colleagues in the R package ConsensusOV[4]. The top panel in **Figure 6** shows the 4 scores used in the ConsensusOV classification on TCGA-OVC. With the intention to automatically provide an understand the OVC biology underlying the transcriptional OVC subgroups through the use of signifinder, the 296 TCGA-OVC samples were studied using the 10 OVC and the 17 pan cancer-specific signatures available in the package. A correlation matrix of signatures' scores was computed (**Figure 3**) to highlight concordant signatures and identify co-occurrences of different processes. The correlation plot clearly shows three groups of signatures. **Group 1** focuses on the extracellular matrix composition and the epithelial-to-mesenchymal transition (EMT) quantifying the mesenchymal-like phenotype. This group highly correlates with the MES samples. **Group 2** is composed of signatures dedicated to many aspects of the immune system in cancer, with two anti-correlated subgroups. The first contains the IMR samples, characterized by chemokine expressions and inflammatory signals, the other containing the PRO samples, known to show low inflammatory response. **Group 3** is dedicated to the signatures of the chromosomal instability and the related mitotic index and cell cycle that reflect the genomic alterations that are widespread and that govern the transcriptional landscape of OVC. The heatmap in **Figure 6** plots the scores of the signatures that most correlate or anti-correlate (correlation over 0.6 or under -0.6) with the MES, PRO and IMR groups. While, ridge plots in **Figure 5** represent the density plot of the scores of signatures in group 1 and 2. From these plots it is even more clear that group 1 characterizes the MES samples and group 2 the IMR and PRO samples with scores going in opposite directions.

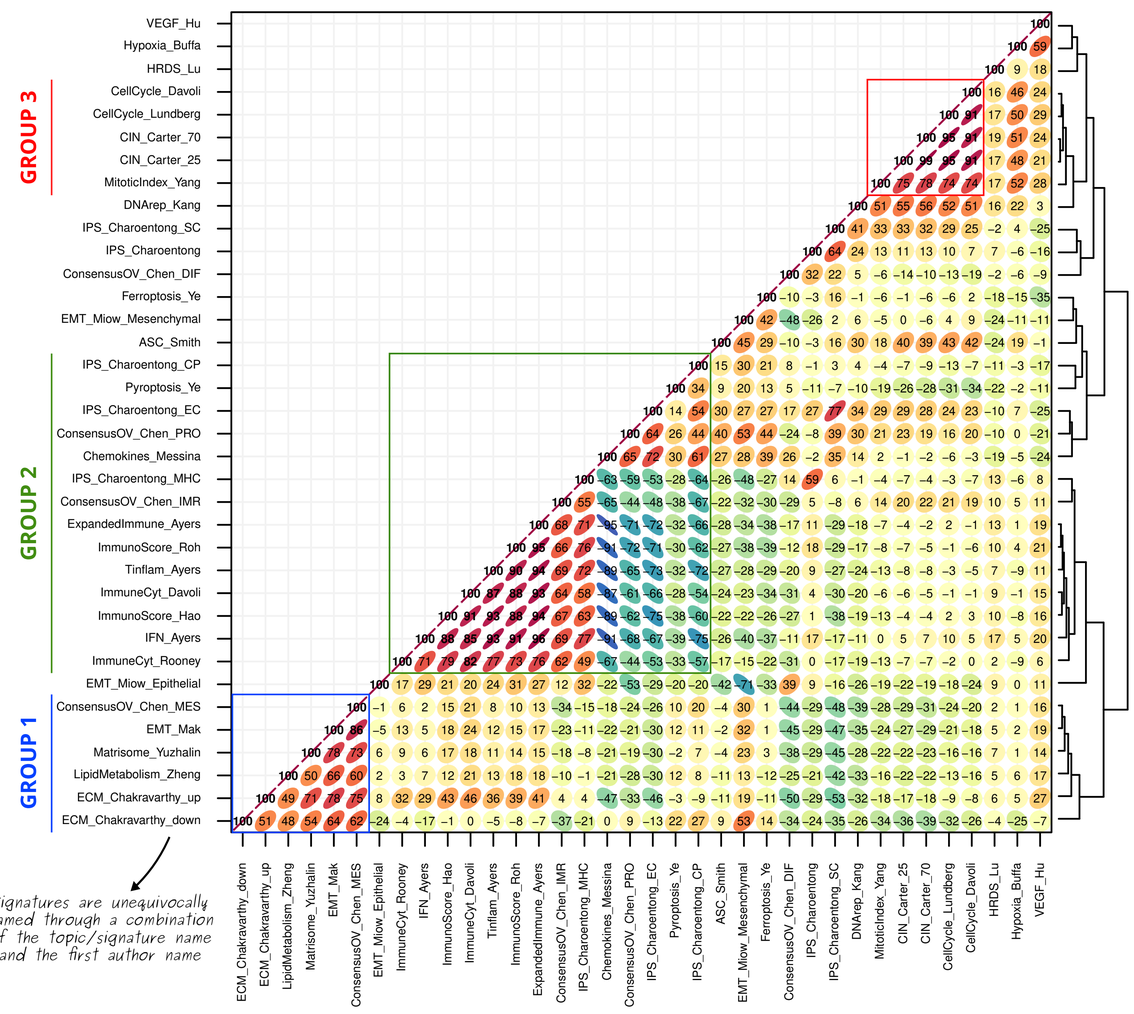

Figure 3 *- correlation plot of the signature scores computed on TCGA-OVC dataset.*

tissue				topic			
pan-tissue	17	kidney	2	immune system	13	metabolism	3
ovary	10	prostate	2	autophagy	4	stem cell	3
brain	3	stomach	2	ferroptosis	4	extracellular matrix	2
lung	3	bladder	1	pyroptosis	4	angiogenesis	1
skin	3	cervix	1	cell cycle	3	carcinoma lesions	1
breast	2	liver	1	chromosomal instability	3	hypoxia	1
colon	2	oral cavity	1	EMT	3	ovarian subtypes	1

23 tumors
13 tissues
14 topics

Figure 4 *- overview of topics and tissues covered by the signatures collected in signifinder.*

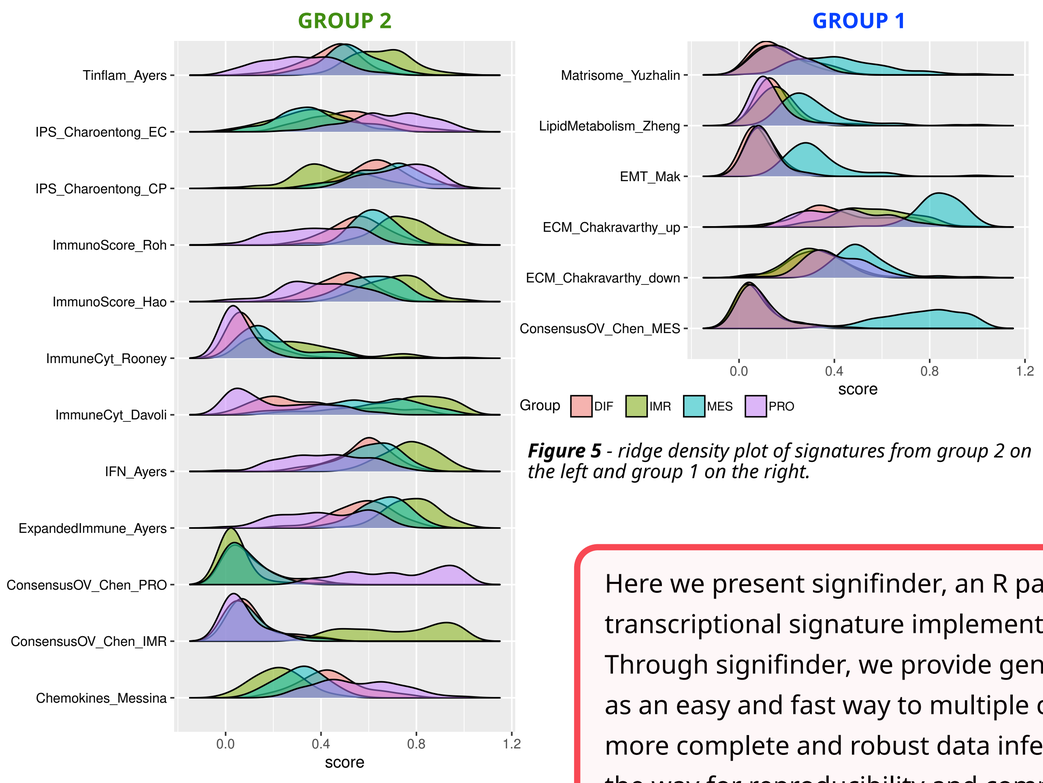

Figure 5 *- ridge density plot of signatures from group 2 on the left and group 1 on the right.*

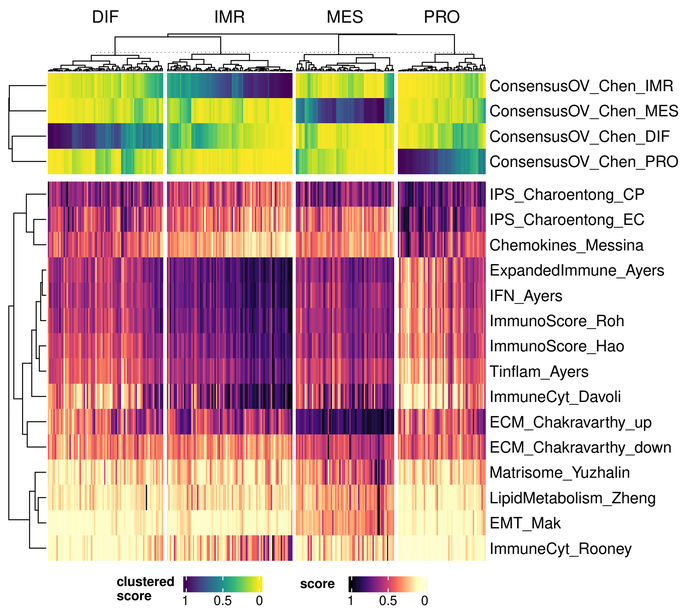

Figure 6 *- Top: heatmap of scores from ConsensusOV. These scores guide the clustering. Bottom: heatmap of signatures that most correlate or anti-correlate with the ConsensusOV scores.*

CONCLUSIONS

Here we present signifinder, an R package dedicated to collect and distribute easy-to-use transcriptional signature implementations helpful to analyze cancer transcriptomes. Through signifinder, we provide general principles for interpreting the signatures, as well as an easy and fast way to multiple cancer signatures computation that would allow for more complete and robust data inferences. We consider signifinder a useful tool to pave the way for reproducibility and comparison of transcriptional signatures in oncology.

Please contact us if you have a signature that you would like to see added to this package. The only requirement is that the signature has publicly available gene list and method to compute it.

The more difficult/custom the implementation, the better, as its inclusion in this package will provide more value for other users in the R/Bioconductor community.

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Bioconductor — *submitted to Bioconductor*

GitHub

freely available on GitHub!

https://github.com/CaluraLab/signifinder

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REFERENCES **1.** K Tomczak, P Czerwińska, and M Wiznerowicz, The Cancer Genome Atlas (TCGA): an immeasurable source of knowledge, Contemp. Oncol. Poznan Pol., vol. 19, no. 1A, pp. A68-77, 2015; **2.** The Cancer Genome Atlas Research Network. Integrated genomic analyses of ovarian carcinoma. Nature 474, 609–615 (2011); **3.** Tothill RW et al. Novel molecular subtypes of serous and endometrioid ovarian cancer linked to clinical outcome. Clin. Cancer Res. 2008; **4.** Chen GM et al. Consensus on Molecular Subtypes of High-Grade Serous Ovarian Carcinoma. Clin Cancer Res. 2018 Oct 15;24(20):5037-5047.